

CUSTOMER EVIDENCE

Name _____ Class _____ Date _____

TASK C1A**Organise the evidence**

A survey asked **60 students**. Students could select both features.

Leakproof: **38**; keeps drinks cold: **34**; both features: **22**.

Each feature total includes students who selected both.

Q1. Sketch a rectangle and two overlapping circles. Label $n(U) = 60$, Leakproof and Keeps cold. Put 22 in the overlap.

Q2. Find the number who selected leakproof only.

Q3. Find the number who selected cold only.

TASK C1B**Complete the evidence**

Use your diagram and results from C1A. The survey total is **60 students**.

Q4. Find the number wanting at least one feature.

Q5. Find the number wanting neither feature.

Q6. Find the percentage wanting at least one feature.

PRODUCT CHOICES**THINK****Which claim is supported?**

Choose **A** or **B**. No calculation.

A 83.3% of these 60 surveyed students wanted at least one feature.

B 83.3% of all Bellerbys students want at least one feature.

The data describe _____ because _____.

TASK C2A**Bottles and lids**

GreenSip offers **4 colours** and **3 lid styles**. A version uses one colour and one lid.

Q1. How many versions are possible?

Q2. If 600 bottles are divided equally, how many of each version?

THINK Does the survey show that every version will be equally popular?

TASK C2B**Choose the method**

Eight students volunteer to test and present GreenSip.

Write the method name first. Then calculate.

Q1. Choose 3 testers. All three have the same role. Does order matter? Choose permutation or combination.

Q2. Appoint a leader, presenter and recorder from the 8. The three roles are different. Does order or position matter?

Optional Choose 3 testers, then choose their leader.

BOTTLE MASS AND UNIT CONVERSIONS**TASK C3A****Find one bottle's mass**

Six identical bottles + a 120 g carton = 1.56 kg total.

Q1. Convert 1.56 kg to grams.

Q2. Let x be one bottle's mass in grams. Write an equation.

Q3. Solve the equation.

Q4. Substitute your value of x to check it.

TASK C3B**Convert the batch mass**

Use your checked mass for one bottle from C3A.

Batch: 600 bottles **Conversion:** $1 \text{ kg} \approx 2.20 \text{ lb}$ **Planning limit:** 330 lb

Q1. Find the total mass in grams.

Q2. Convert grams to kilograms.

Q3. Convert kilograms to pounds.

Q4. Compare the result with the 330 lb limit.

PACKAGING: QUADRATIC EQUATION**TASK C3C****Rectangular insert**

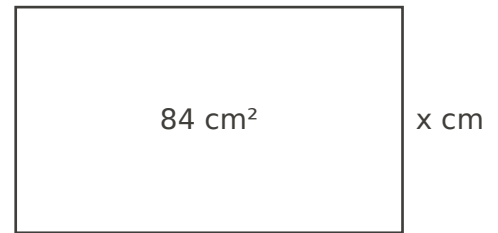
A recycled-card insert is a rectangle.

Width = x cm

Length = $(x + 5)$ cm

Area = 84 cm^2

$(x + 5) \text{ cm}$



Q1. Write the area equation.

Q2. Write it in standard quadratic form.

Q3. Solve by factorising.

Q4. Choose valid dimensions and check.

INPUT MATERIAL AND WASTE

TASK C4A

Allow for material loss

Finished bottle mass = 144 kg.

Ten per cent of the **INPUT material** is lost.

Q1. What percentage of the input remains?

Q2. Let M be the input mass. Write an equation.

Q3. Find the required input mass and the waste.

Q4. Reverse-check your answer.

MATERIAL MIXTURE AND ORDERS

TASK C4B

Complete the 3:2 mixture

The **160 kg input** is the combined total of Materials A and B.

Required ratio A:B = 3:2. Available now: **90 kg of A.**

Q1. Using 90 kg of A, find one provisional ratio part.

Q2. Find the amount of B that matches this 90 kg of A.

Q3. Find the provisional total and its shortage from 160 kg.

Optional extra step Divide the shortage in the ratio 3:2.

TASK C5A

Orders and discounts

GreenSip has **600 confirmed one-bottle orders.**

Assume all are paid at the stated discounted prices.

Student:staff = 4:1. List price = **¥80.**

Student discount = **15%**; staff discount = **10%.**

Q1. Find the number of student and staff orders.

Q2. Find the student discount amount and final price.

Q3. Find the staff discount amount and final price.

MONEY RECEIVED AND EVIDENCE**TASK C5B****¥40,000 target**

Use the confirmed one-bottle orders and stated discounted prices.
Complete the money-received cells and compare the total with the target.

Group	Number	Price each	Money received
Q1 Students	480	¥68	
Q2 Staff	120	¥72	
Q3 TOTAL	600	-	
TARGET	-	-	¥40,000
Q4 COMPARE	-	-	Total - target = _____

WORKING FOR Q1-Q4

Optional By what percentage is the total above the ¥40,000 target?

THINK**Which claims are supported?**

For each statement, write **SUPPORTED** or **NOT SUPPORTED** and give a reason. Use evidence from the case.

A These 600 confirmed orders collect your calculated total at the stated discounted prices.

B Every future GreenSip batch will collect that same total.

C GreenSip should make 50 of every colour-and-lid version.

Claim	Supported / Not supported	Reason
A		
B		
C		

INDIVIDUAL EXIT CHECK

TASK C6

EcoCup: apply the Week 1 toolkit

EcoCup plans **240 cups** using **3 colours** and **2 lids**.

Finished material: 84 kg **Input loss:** 20% **Material ratio:** A:B = 2:1

Work independently. Show calculations and units.

Q1. How many versions are possible?

Q2. If divided equally, how many cups per version?

Q3. Find the required material input.

Q4. Find the quantities of A and B.

Q5. State one assumption that still needs evidence.